

9(a)	any nine from: 1 DNA cut with restriction enzymes ; 2 (DNA) fragments placed in wells in gel ; 3 at cathode ; 4 current / electric field, applied ; 5 fragments negatively charged ; 6 move towards anode ; 7 gel acts as a molecular sieve ; 8 smaller fragments move, faster / further, than larger ones ; 9 current switched off ; 10 <i>ref. to</i> Southern blotting / AW ; 11 <i>ref. to</i> staining / (gene / DNA) probes, for visualisation ; 12 alleles have different positions on gel ; 13 AVP ; e.g. desired, DNA / allele, selected / increased, by PCR	9
------	---	---

Question	Answer	Marks
9(b)	any six from: <i>if present</i> 1 enables early treatment ; 2 lifestyle changes ; 3 elective / preventative, mastectomy ; 4 regular check ups ; 5 prevents unnecessary prolonged suffering if discovered and treated early ; 6 prevents early death ; 7 AVP ; e.g. social or family advantage of preventing parent's early death <i>if not present</i> 8 removes worry ; 9 <i>ref. to</i> planning a family ; 10 AVP ; e.g. prevention / early diagnosis, cheaper than later treatment	6